

```
(%o5) (y+x)^2-2*y^2+y*x
(%i6) string(expand(p1 + p2 + p3));
(%o6) -y^2+2*x*y+y+x^2+x
(%i7) string(expand(p3 - p1));
(%o7) 2*y^2+2*x*y-y
(%i8) string(expand(p1 * p2));
(%o8) y^4-y^3-x*y^2-x^2*y+x*y-x^4+x^3
(%i9) string(p1 / p2);
(%o9) (-y^2+y*x^2)/(-y^2-x^2+x)
(%i10) string(divide(p1, p2));
(%o10) [1,y+2*x^2-x]
(%i11) string(divide(x^2 - y^2, x + y));
(%o11) [x-y,0]
(%i12) quit();
```

Note that the `/` operator just places the polynomials as fractions, rather than dividing them. The function `divide()` actually divides the first polynomial by the second one, yielding a list with the quotient and the remainder of the division. If the division needs to be in relation to a particular variable, that can be passed as the third argument to divide. Check out the variation below, to understand what this means:

```
$ maxima -q
(%i1) string(divide(x^2 - y^2 + y, x + y, x));
(%o1) [x-y,y]
(%i2) string(divide(x^2 - y^2 + y, x + y, y));
(%o2) [-y+x+1,-x]
(%i3) quit();
```

Coefficients of a polynomial

Extracting the coefficients of a polynomial is another basic and common requirement for polynomial manipulations. Maxima provides two slightly different mechanisms. The first one just finds the coefficient of a given variable or its power, using `coeff()`. The second one segregates a polynomial into the coefficient of a given variable or its power, and the remaining terms, using `bothcoef()`.

```
$ maxima -q
(%i1) string(bothcoef(expand(x^2 - y^2 + (x + y)^2), x^2));
(%o1) 2,2*x*y
(%i2) string(bothcoef(expand(x^2 - y^2 + (x + y)^2), x));
(%o2) [2*y,2*x^2]
(%i3) string(bothcoef(expand(x^2 - y^2 + (x + y)^2), y^2));
(%o3) [0,2*x*y+2*x^2]
(%i4) string(bothcoef(expand(x^2 - y^2 + (x + y)^2), y));
(%o4) [2*x,2*x^2]
(%i5) string(coeff(expand((x + 2 * y)^50), x^20));
(%o5) 50604606318512743383040*y^30
(%i6) string(coeff(expand((a + b + c + d)^4), a^3));
(%o6) 4*d+4*c+4*b
(%i7) quit();
```

Polynomial fractions

Calculating the greatest common divisor (GCD) is one of the very useful operations to simplify the fractions of polynomials. Other common requirements are extracting the numerator, the denominator, and the highest power. Here is the function demonstrating all of these:

```
$ maxima -q
(%i1) gcd(x^3 + 3*x^2 + 3*x + 1, x^2 + 3*x + 2);
(%o1) x + 1
(%i2) string(ezgcd(x^3 + 3*x^2 + 3*x + 1, x^2 + 3*x + 2));
(%o2) [x+1,x^2+2*x+1,x+2]
(%i3) string(denom((x + 1)^-3 * (1 - x)^2));
(%o3) (x+1)^3
(%i4) string(num((x + 1)^-3 * (1 - x)^2));
(%o4) (1-x)^2
(%i5) hipow(expand((x + 1)^3 + (1 - x)^3), x);
(%o5) 2
(%i6) quit();
```

Note that the `ezgcd()` function lists out the remainder polynomials, along with the GCD.

Polynomial fractions can be differentiated using the powerful `ratdiff()`:

```
$ maxima -q
(%i1) string(ratdiff((x + 1)^-1 * (1 - x)^2, x));
(%o1) (x^2+2*x-3)/(x^2+2*x+1)
(%i2) string(ratdiff(1 / (x + 1), x));
(%o2) -1/(x^2+2*x+1)
(%i3) string(ratdiff((x^2 - 1) / (x + 1), x));
(%o3) 1
(%i4) quit();
```

And `ratsubst()` is a powerful expression substitution function, with intelligence. It can dig into the expression to simplify complicated expressions, including trigonometric ones. Check out the `%i5`, for one of its powerful applications. `ratsubst(<new>, <old>, <expr>)` replaces the `<old>` expression by the `<new>` expression in the complete expression `<expr>`:

```
$ maxima -q
(%i1) string(ratsubst(u, x^2, x^3 + 3*x^2 + 3*x + 1));
(%o1) (u+3)*x^3+u+1
(%i2) string(ratsubst(u, x^2, (x+1)^3));
(%o2) (u+3)*x^3+u+1
(%i3) string(ratsubst(u, x^2, (x+1)^4));
(%o3) (4*u+4)*x*u^2+6*u+1
(%i4) string(ratsubst(u, x - 1, x^4 - 2*x^2 + 1));
(%o4) u^4+4*u^3+4*u^2
(%i5) string(ratsubst(sin(x)^2, 1 - cos(x)^2, cos(x)^4 - 2*cos(x)^2 + 1));
(%o5) sin(x)^4
(%i6) quit();
```

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